

Shubhankar Varshney

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About Me

I am a graduate student in Computer Science Department of Purdue University. I am interested in problems which lie in the intersection of multiple fields like Computer Science, Mathematics, Biology, Economics, etc. My current research interest is in Computational Topology, specifically I am working on multiparameter persistence.

Education

Purdue University, West Lafayette

PhD Computer Science

August 2024 – Present

Chennai Mathematical Institute, Chennai

M.Sc. Theoretical Computer Science

Master's Thesis Topic: Making sense of Sample Points in Euclidean Domain

Advisor: Prof. Priyavrat C Deshpande

August 2022 – Jun 2024

CGPA: 9.12/10

Indian Statistical Institute, Bangalore

Bachelor of Mathematics (Hons.)

July 2019 – May 2022

90% (First Division with Distinction)

Courses Undertaken

Mathematics

- Group, Ring and Field Theory
- Linear Algebra
- Galois Theory
- Graph Theory
- Optimization
- Topology
- Real and Complex Analysis
- Differential Geometry
- Probability Theory
- Statistics
- Algebraic Topology
- Finite Field Combinatorics
- Enumerative Combinatorics
- Mathematical Logic

Computer Science

- Theory of Computation
- Computational Geometry
- Advanced Algorithms
- Concurrent Programming
- Topological Data Analysis
- Infinite State Verification
- Formal Security Analysis
- Algorithmic Coding Theory
- Design and Analysis of Algorithms
- Algorithmic Game Theory
- Programming Language Concepts
- TCS Toolkit

Internships

Verification of Event-Driven Sequentially Consistent Memory Models

May 2023 - August 2023

Uppsala University, Sweden

- Worked under the guidance of Prof. Mohamed Faouzi Atig, Prof. Parosh Aziz Abdulla and Prof. Ramanathan Thinniyam.
- Defined syntax and semantics of Event-driven model and proved complexity results for some models which were made by adding different constraints.
- Work is ongoing to publish a paper on it.

Efficient Transmission of TCU Data with Special emphasis on GPU data

May 2022 - July 2022

Maruti Suzuki India Limited

- Worked on Telematics data to make driving score which could be used for usage-based insurance (UBI).
- Then developed an algorithm to reduce the bandwidth used for transmission of GPS data with almost no compromise with precision and accuracy.
- Deployed the algorithm on an embedded system based on Yocto Project sumo 4.14.

Projects

Multiparameter Persistence

January 2025

Working on making efficient algorithms to compute topological invariants in multiparameter setting.

Kernel method for Persistence Diagrams via kernel embedding and weight factor

November 2023

This was done as part of Topological Data Analysis Course. A paper with the same name was presented. In the paper a new kernel method was proposed for persistence diagrams which had many stability properties which other kernel methods lacked. Experiments were shown that pointed to the fact that this method had better classification rates than other methods.

Decoration of Shi Threshold Arrangements

December 2022

This was done as part of Enumerative Combinatorics course. The problem was to calculate the number of regions formed by Shi hyperplanes in d -dimension. A bijection was constructed from the regions to a decorated permutation of d elements, where decorated means that some more information is provided in addition of the permutation. I used arcs between 2 elements in the permutation as decorations.

Winter and Power Trends in South Korea

October 2020

This was done as a group project in Statistics-I course. We plotted quantile graphs and did quantile regression on total electric power load vs temperature. Here we found sample bias in the paper that we had to present. The report can be found [here](#).

Academic Achievements

TIFR Entrance Exam

2022

Got selected for interview in TIFR GS2022-IPhd CS admissions.

Teacher's Appreciation Contingency Grant

2019

Got contingency grant of 2000 INR for books because I came 3rd in B.Math first semester.

Regional Mathematical Olympiad (RMO)

2018

Cleared RMO 2018 for XII class students being among the top 6 students in the Uttar Pradesh state.

Merit Scholarship

2017

Was among 20 students out of 3000 of Aligarh Muslim University to get selected for Scholarship for the duration of 11th – 12th standard based on scholarship test.

Teaching Experience

C Programming

Jan 2026 – May 2026

○ Teaching Assistant (TA) for the CS240 C-Programming course. (Instructor for the course was Prof. Turkstra)

Theory of Computation

Sept 2023 – December 2023

○ Teaching Assistant (TA) for the Theory of Computation course for B.Sc 2nd year and M.Sc CS 1st year students in CMI. (Instructor for the course was Prof. V. Arvind)

Limit Camp Mentor

June 2021 – July 2021

○ Limit is a test organised by students of ISI Bangalore. I was one of the mentors for students who got selected for the Limit camp.

Seminars Attended

ICERM Workshop on Geometry and Topology

2026

Attended workshop on Topology and Geometry in Rhode Island, Providence.

Graduate student conference in topology and geometry

2026

Gave a talk about Introduction to Computational Topology in Rutgers.

Workshop on Applied Topology and Complex Networks

2024

There were two workshops in CMI, Chennai. One was on using Borsuk-Ulam Theorem to prove multiple results and the other entailed a survey for methods and algorithms for temporal networks.

Talks

Mathematical Foundations of Computational Topology	2026
Seminar talk in CMI, Chennai	
Tackling Multiparameter Persistent Homology: Computing Minimal Presentations 2,3 and beyond 2026	
Seminar Talk in CMI, Chennai	
Introduction to Computational Topology	2026
Math Club, ISI, Bangalore	
Introduction to Computational Topology	2026
GSTGC, Rutgers, New Brunswick	

Grants and Scholarships

Grant to Attend ICERM, Providence	2026
Grant to Attend and give a talk in GSTGC, Rutgers, New Brunswick	2026
Chennai Mathematical Institute Stipend for MSc CS	2022-24

Technical Skills

- **Programming languages:** C, C++, Python, Haskell, Java. **Tools:** $\text{\LaTeX}$